Real-to-Sim-to-Real at the Last Millimeter

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Abstract: We study drawer and charger insertion with 0.7–2 mm clearances on a roughly $1,000 arm with high joint friction and unmodeled joint play through real-to-sim-to-real learning. Simulators constructed from a few real demonstrations supply state-based teachers for image-based students. The students complete 12/20 drawer trials and 18/20 charger trials on the real arm. Three lessons emerge. First, directly revising scripted teachers shortened development and made simulator-specific behaviors easier to correct. Second, smoothing decision boundaries and matching action chunks to student execution make supervision more consistent. Third, joint PD control with commanded and measured joint positions supports contact-guided insertion. Our experience supports scripted teachers as a practical starting point, with learnability and contact feedback central to transfer. Video: https://ripl.github.io/agent_written_teachers.

Keywords: Real-to-sim-to-real, Precise insertion, Teacher-student distillation

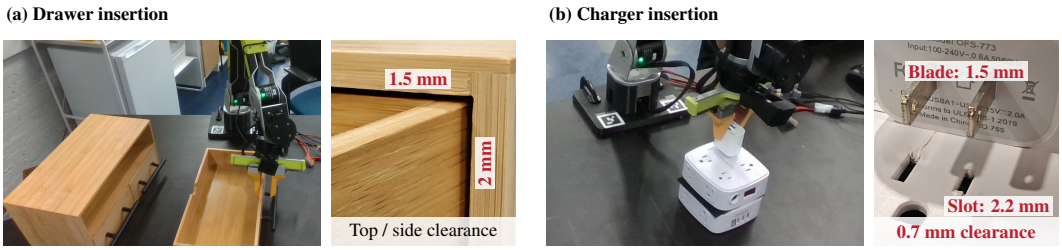


Figure 1: Real drawer (a) and charger (b) insertion on a roughly $1,000 arm, viewed from side cameras. Insets show separate close-up photographs with measured dimensions.

Introduction. At millimeter clearances, slight misalignment can jam an insertion. Prior demonstrations include RebarSim on a Franka FR3 [1] and ALOHA on roughly $5,000 arms [2]. Our arm costs roughly $1,000, and despite substantial system identification, our simulator cannot match it to within the clearance. The arm usually repeats a motion within 1 mm, but friction stops its joints up to 7° short of their targets. Our best rigid model still misplaces the wrist camera by a median 10 mm, suggesting the arm is not rigid (Fig. 2b). Even within simulation, replaying the same action sequence on GPU instead of CPU, or with a smaller timestep, shifts the final drawer position by 1–2 cm (Fig. 2a). Policies must therefore tolerate these errors. We report three lessons from training such policies with scripted teachers in simulation.

Tasks and pipeline. A 6-DoF Seeed reBot B601-DM picks a loose drawer and inserts it into a free-standing cabinet, or transfers a phone charger between adjacent outlets (Fig. 1). The Isaac Sim drawer model is fitted to ten teleoperated demonstrations. The charger model is built from one, with geometry measured in its images. Training randomizes masses, frictions and grip force.

In simulation, a teacher sees the simulator state and labels actions for a student. Prior work usually trains the teacher with reinforcement learning (RL) [1, 3, 4, 5, 6]. Our teachers are instead Python

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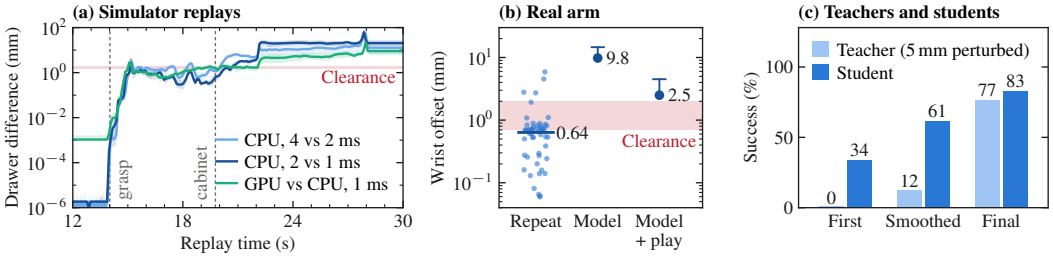


Figure 2: (a) Drawer-position differences between replays of the same action sequence at different timesteps or on GPU versus CPU (median and interquartile range, 64 paired starts). (b) Real wrist offsets across repeats, and model error (median, maximum). (c) Successive scripted teachers under pose perturbations and their students. Panels (a) and (c) show simulation results.

programs written and revised by coding agents [7]. The student sees scene and wrist images plus commanded and measured joint positions, and outputs action chunks [2]. We train it with online DAgger [8], in which the teacher labels the states the student visits.

1. Direct teacher revision shortens development. We spent roughly two weeks developing RL teachers and two days developing scripts, with coding agents assisting both. Sparse demonstrations leave contact behavior underdetermined [9]. Our RL teachers learned edge grasps that relied on cabinet support: in simulated free-hanging tests, they dropped the drawer in 11–12 of 12 trials, versus 1 of 12 for centre grasps. Scripts let us revise grasps and recovery directly without retraining a teacher.

2. Successful scripts must also be learnable. ExpertGen improves students by refining scripted teachers with RL [10]. We instead revise the programs themselves. An early student reached 90.3% simulated success yet repeatedly opened and closed its gripper at the real drawer. Sharp teacher branches assigned conflicting commands to nearby states that images may not distinguish. We perturb object poses across 16 copies by 5 mm and 1°, then execute the median joint action and majority gripper vote. Smoothing reduced gripper disagreement at closing from 28.6% to 0.4%. Across three revisions, all teachers exceeded 90% success on exact poses, but perturbed success better tracked student performance (Fig. 2c). For action chunks, we generate labels from the teacher's own plan over the student's execution horizon, keeping supervision consistent with chunked deployment.

3. Use contact to resolve alignment uncertainty. Under joint proportional–derivative (PD) control, contact can prevent the arm from reaching its target. Without torque sensors, light contact is hard to tell from friction. We supply both commanded joints, q_{des}, and measured joints, q, making $q_{\mathrm{des}} - q$ available as a contact-resistance cue. The drawer teacher originally retreated on millimeter pose thresholds available only in simulation. It now uses commanded–measured motion to retreat, realign, and push again, basing recovery on information the deployed student also receives [11].

Real results and implications. The students complete 12/20 drawer trials and 18/20 charger trials (Table 1). Drawer completion includes the final push to close. Charger completion includes insertion and release. Recorded drawer failures include successful entry followed by prolonged holding without releasing for the final push. The matched RL-student comparison remains in progress. Our experience favors developing learnable, contact-aware scripted teachers before RL refinement.

Task	Successful real trials / total
Drawer (including final push)	12/20 (60%)
Charger (insert and release)	18/20 (90%)

Table 1: Real-world success of students distilled from scripted teachers.

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